INCH-POUND

MIL-DTL-43375F
01 August 2023
SUPERSEDING
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18 September 1990

DETAIL SPECIFICATION

CLOTH, DUCK, NYLON, 12.5 OUNCE

This document is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1. SCOPE

1.1 Scope. This specification covers 12.5 ounce, nylon, duck cloth.

1.2 Classification. The cloth will be of the following classes as specified (see 6.2).

1.2.1 Classes.

Class 1 - Dyed
Class 2 - Dyed, water repellent

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-3064 - Evaluation of Quality of Textile Materials

(Copies of this document are available online at https://quicksearch.dla.mil)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

OTHER GOVERNMENT DOCUMENTS AND PUBLICATIONS

CODE OF FEDERAL REGULATIONS

Title 16 CFR Part 303 - Rules and Regulations Under the Textile Fiber Product Identification Act
Title 40 CFR Parts (150-180) – Protection of Environment

(Copies of these documents are available online at https://www.ecfr.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP1 - Evaluation Procedure for Gray Scale for Color Changes
AATCC EP2 - Evaluation Procedure for Gray Scale for Staining
AATCC EP8 - Evaluation Procedure for AATCC 9-Step Chromatic Transference Scale
AATCC EP9 - Evaluation Procedure for Visual Assessment of Color Difference of Textiles
AATCC TM8 - Test Method for Colorfastness to Crocking: Crockmeter
AATCC TM16.3 - Test Method for Colorfastness to Light: Xenon Arc
AATCC TM20 - Test Method for Fiber Analysis: Qualitative
AATCC TM22 - Test Method for Water Repellency: Spray
AATCC TM61 - Test Method for Colorfastness to Laundering: Accelerated
AATCC TM81 - Test Method for pH of the Water-Extract from Wet Processed Textiles
AATCC TM135 - Test Method for Dimensional Changes of Fabrics after Home Laundering

(Copies of these documents are available online at https://aatcc.org.)

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AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM D737	- Standard Test Method for Air Permeability of Textile Fabrics
ASTM D3775	- Standard Test Method for End (Warp) and Pick (Filling) Count of Woven Fabrics
ASTM D3776/D3776M	- Standard Test Method for Mass Per Unit Area (Weight) of Fabric
ASTM D5034	- Standard Test Method for Breaking Strength and Elongation of Textile Fabrics (Grab Test)

(Copies of these documents are available online at https://www.astm.org.)

OTHER PUBLICATIONS

SDL ATLAS

Part Number 402985 – Slub/Knot Replica Set

(Replica set is available for purchase from SDL Atlas Customer Service, 1-803-329-2110.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.1.

3.2 Standard sample. Unless otherwise indicated, the finished cloth shall match the standard sample for shade and appearance, and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

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3.4 Material.

3.4.1 Yarn. The yarn for the warp shall be 840 nominal denier, bright, high tenacity, multifilament nylon and the yarn for the filling shall be 420 nominal denier, 3 ply or single equivalent denier, bright, high tenacity, multifilament nylon when tested as specified in 4.5.

3.4.2 Weave. The cloth shall be woven using a plain weave, one (1) up, one (1) down, when tested as specified in 4.5.

3.5 Color. The color of the finished cloth shall be as specified in the contract or procurement documents, (see 6.2).

3.5.1 Visual shade matching. The color the finished cloth shall match the standard sample when tested as specified in 4.6.

3.5.2 Colorfastness. Unless otherwise specified in the contractor procurement documents, the finished cloth referenced in 3.4 and 3.5 shall conform to the colorfastness requirements listed below in Table I when tested as specified in 4.5.

TABLE I. Colorfastness requirements.

Colors Evaluation	Laundering Color Change and Staining, grade, (min.)	Light (after 40 AFU or 170 kJ/(m^2nm) @420 nm), grade, (min.) 1/	Crocking Dry/Wet, grade, (min.)
All colors	3.5	3.5	3.5

1/ AFU: AATCC Fading Units

3.6 Spectral reflectance.

3.6.1 Camouflage Green 483. When specified, Camouflage Green 483 shall conform to the reflectance values listed below, in Table II, when tested as specified in 4.6.3.

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TABLE II. Spectral reflectance requirements.

Reflectance values (percent)		
Wavelength, Nanometers (nm)	Camouflage Green 483	
	Min.	Max.
600	3	10
620	3	10
640	3	10
660	3	11
680	3	15
700	4	28
720	5	40
740	7	52
760	11	60
780	17	64
800	24	67
820	32	70
840	37	71
860	40	73

3.7 Physical requirements. The finished cloth shall conform to the requirements, listed below, in Table III, when tested as specified in 4.5.

TABLE III. Physical requirements.

Characteristic	Requirement
Weight, oz./sq.yd. (min.)	12.5
Yarns per inch, (min.) Warp Filling	 56 28
Breaking strength, pounds, (min.) Warp Filling	 800 700
Air permeability, cu.ft/min./sq.ft, (max.)	3.0

3.8 Finish. The Class 1 cloth shall be desized, scoured, dyed and heat set. The Class 2 cloth shall be desized, scoured, dyed, heat set, and water repellent treated. The cloth shall not be bleached in any manner of the process.

3.8.1 Water repellency, Class 2. The Class 2 cloth shall be given a water repellent treatment on both sides of cloth. The cloth shall meet a minimum spray rating of 90, 90, 80 when tested as specified in 4.5.

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3.9 pH. The pH of the water extract of the finished cloth shall be no less than 5.5 nor more than 8.5 when tested as specified in 4.5.

3.10 Dimensional stability. The shrinkage or elongation both in the warp and filling of the finished cloth shall have an average dimensional change of 3.5% maximum in either direction. Testing shall be as specified in 4.5.

3.11 Toxicity. Unless otherwise specified in the contract or procurement documents, the finished cloth shall not present a health hazard and shall show compatibility with prolonged, direct skin contact when tested as specified in 4.7.

3.12 Face identification. The face of the finished cloth shall be identified by stamping the word “FACE” on that side at each end of the cloth.

3.13 Width. For Government procurements only, the width of the finished cloth shall be as specified in the contract or purchase order (see 6.2) and shall be the minimum acceptable width exclusive of the selvage. Woven or heat slit selvages shall be used.

3.14 Length and put-up. For Government procurements only, unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each roll not less than 80 yards. Each length shall be put-up full width on a roll as specified in 5.1.

3.15 Fiber identification. Each roll of finished cloth shall be labeled or ticketed for the fiber content in accordance with the Title 16 CFR Part 303 Rules and Regulations under the Textile Fiber Products Identification Act.

3.16 Workmanship. The finished cloth shall be uniform in quality and shall conform to the quality of product established on this specification. The occurrence of defects as specified in 4.4, shall not exceed the acceptance quality levels as specified in the contract or purchase order.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3.1)
b. Conformance inspection (see 4.3.2)

4.2 Inspection conditions. Unless otherwise specified, excluded, amended, modified or qualified in this specification or applicable procurement documents (see 6.2), all inspections shall be performed in accordance with all the requirements of referenced documents.

4.3 Types of inspections.

4.3.1 First article inspection. The first article, submitted in accordance with 3.1 shall be inspected for appearance, color, the defects of 4.4 and tested for the characteristics as specified in 4.5 through 4.7.

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4.3.2 Conformance inspection. Conformance inspection shall include the examination of 4.4 and the tests of 4.5 through 4.7 as applicable. Sampling for inspection shall be performed in accordance with ASQ/ANSI ZI.4 and with acceptance quality limits as specified in the contract and/or procurement document, except where otherwise indicated (see 6.2).

4.4 Visual examination. Each roll in the sample shall be examined yard-by-yard on the face side for defects in accordance with Table IV and Type I of MIL-STD-3064 except for slubs and knots which shall be in accordance with Table IV.

TABLE IV. Defects descriptions.

Creases in the fabric
Wrong draw, appearing as a line in the warp direction for more than 18" in length
Slubs greater than 1-1/2" long and greater than 1/16" wide
Knots 1/8" or greater

4.4.1 Roll identification and marking (face/back) examination. During the yard-by-yard examination, each roll in the sample shall be examined for defects as specified in MIL-STD-3064.

4.4.2 Shade variation examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade variation as specified in MIL-STD-3064.

4.4.3 Length examination. For Government procurements only, during the yard-by-yard examination, each roll in the sample shall be examined for length as specified in MIL-STD-3064.

4.5 End item testing. The cloth shall be tested for the characteristics listed in Table V. The methods of testing as specified wherever applicable and as listed in Table V shall be followed. All test reports shall the individual values utilized in expressing the final results. The sample unit shall be five (5) continuous yards full width of the finished cloth for all physical and chemical tests. The lot shall be unacceptable if one (1) or more tests fail to meet the requirements specified. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE V. Material testing.

Characteristic	Requirement Reference	Test method
Fiber identification	See 3.4.1	AATCC TM20 (see 6.4)
Weave	See 3.4.2	Visual
Visual shade matching	See 3.5.1	See 4.6.1
Colorfastness: Laundering Light Crocking	 See Table I See Table I See Table I	 AATCC TM61 Test 1A, 1/ AATCC TM16.3, Option 3, 2/ AATCC TM8, 3/
Spectral reflectance	See Table II	See 4.6.3
Weight	See Table III	ASTM D3776/D3776M
Yarns per inch	See Table III	ASTM D3775
Breaking strength	See Table III	ASTM D5034
Air permeability	See Table III	ASTM D737
Spray rating	See 3.8.1	AATCC TM22
pH	See 3.9	AATCC TM81
Dimensional change	See 3.10	AATCC TM135, (3)(V)(A)
Toxicity	See 3.11	See 4.7

1/ Rated using the AATCC EP1 and AATCC EP2
2/ Rated using the AATCC EP1
3/ Rated using the AATCC EP8

4.6 Methods of testing and inspection.

4.6.1 Visual shade matching. The color and appearance of the cloth shall match the standard sample when viewed using the AATCC EP 9, Option C, (see 6.5) with a primary light source simulating the spectral quality of average daylight, CIE Illuminant D65, with a color temperature of 6500 (± 200) Kelvin (K) illumination of 100 (± 20) foot candles. Alternatively, the color and appearance of the cloth shall match the standard sample with a primary light source simulating artificial daylight, CIE Illuminant D75, with a color temperature of 7500K (± 200) illumination of 100 (± 20) foot candles in lieu of D65. (See 6.5.1). The cloth shall also be a good match to the standard sample with a secondary source simulating the spectral quality of incandescent lamplight, CIE Illuminant A, with a color temperature of 2856K (± 200).

4.6.2 Colorfastness evaluation testing of printed cloth. When testing for colorfastness properties on printed cloth, each color shall be evaluated, whenever possible, separately and reported as such. In cases where the print pattern does not allow for the evaluation of each color separately, the test results should indicate which colors were evaluated together.

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4.6.3 Spectral reflectance test. Spectral reflectance data shall be obtained from 600 to 860 nanometers (nm) for solid colors at 20 nm intervals on a spectrophotometer relative to the polytetrafluoroethylene (PTFE) family of compounds, the preferred white standard. Other white reference materials may be used provided they are calibrated to absolute white or vitrolite tiles. The spectral band width shall be less than 20 nm at 860 nm. Reflectance measurements shall be made by either the monochromatic or polychromatic mode of operation. When the polychromatic mode of operation is used, the spectrophotometer shall operate with the specimen diffusely illuminated with the full emission of a continuous source that simulates either CIE Source A or CIE Source D65. Measurements shall be taken on a minimum of two (2) different areas and the data averaged. The measured areas should be at least 6-inches away from the selvage. The specimen shall be measured as a single layer backed with three (3) layers of the same fabric and shade. The specimen shall be viewed at an angle no greater than 10° from normal, with the specular component included. Specimens shall be oriented in different directions during testing. When possible, the specimens tested shall not contain the same warp or filling yarns when presented to the sample port. Always use the largest aperture possible. Photometric accuracy of the spectrophotometer shall be within one (1) percent and wavelength accuracy within two (2) nm. Any color having spectral reflectance values falling outside the limits at four (4) or more of the wavelengths specified shall be considered a test failure.

4.7 Toxicity test. When the toxicity requirement (see 3.11) can be demonstrated with historical use data on the finishing treatments used, toxicity testing may not be required, (see 6.4 and 6.6).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Department or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The cloth is intended for use in collapsible canteen covers and other equipage items.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Class required (see 1.2).
c. The specific issue of individual documents referenced (see 2.1).
d. When first article is required (see 3.1).
e. Color required (see 3.5).
f. When toxicity is required (see 3.11)
g. Width of cloth required (see 3.13)
h. Length required if other than specified (see 3.14).
i. Inspection conditions (see 4.2).
j. Conformance inspection acceptance quality limits (AQLs) (see 4.3.2).
k. Packaging (see 5.1).

6.3 Standard sample. For access to standard samples and pattern drawings, address the contracting activity issuing the invitation for bids or request for proposals.

6.4 Certificate of compliance. The contracting activity may select to accept a certificate of compliance for stated requirement.

6.5 Visual shade matching. In 2017, Option A of AATCC EP9 was changed to Option C. NOTE: In case of confusion, the viewing geometry should be "The specimen plane and illumination source will be parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35 (± 5º) angle relative to the horizontal. The observer will view the specimens at a 90º angle, relative to the plane of the specimens".

6.5.1 In 2022, the U.S. Military (All Services) accepted D65 as the new preferred light source for visual shade matching due to the supply chain issues with D75.

6.6 Toxicity testing. If dermal toxicity testing is required, it will be conducted using Title 40, Protection of Environment, Code of Federal Regulations current edition and applicable sections, which are also consistent with the following OECD tests:

Skin Sensitization 406 and Dermal Irritation 404 (https://www.oecd-ilibrary.org/environment/oecd-guidelines-for-the-testing-of-chemicals-section-4-health-effects_20745788?page=1)
Dermal Sensitization (https://www.law.cornell.edu/cfr/text/40/798.4100)
Dermal Toxicity (https://www.law.cornell.edu/cfr/text/40/798.2250)

6.7 Subject term (key word) listing.

Canteen
Cover
Equipage
Water repellent treated

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6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - GL

Preparing activity:
DLA-CT

Review activities:
Army - MD

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